Ans-1 2 [B] (2)

Ans-2 Femur [C] (2)

Ans-3 closing of Suture [A] (2)

Ans-4 Male skull [A] (2)

Ans-5 Identifying Individuals [A] (2)

Ans-6 Head shape [C] (2)

Ans-7 facial Recognition [D] (2)

Ans-8 All of the above [D] (2)

Ans-9 Age at death [A] (2)

Ans-10 They are unique to each individual [A] (2)

Ans-
11.1

(5)

Ossification :-

Ossification is defined as form process of formation of bone starting from cartilage or connective tissue to fully developed bone.

Type of Ossification :-

1] Intramembranous Ossification
2] Endochondrial Ossification.

* Endochondrial Ossification :-

In Endochrondrial Ossification bone is developed from Cartilage which is replaced by bone after a period of time.

- In Endochondrial Ossification Cartilage model is necessary.
- The Enchochodrial Ossification is developed in bone which is catagorized by basis of development.

- Endochondrial Ossification is seen in Pelvis & long bone.

* Process of Endochondrial Ossification :-

- Endrochondroial Ossification starts from Cartilage or Connective tissue at 8th week of Intra-utrine life then starts forming further more before become a bone in Cartilage phase it induces a line which is used for the determination of the Individual, then after becoming the Cartildege bone the Cartilage is replaced by bone thr in Endochondrial Ossification specially in Cartilage bone

Ex Pelvis, long bone in Cartilage bone.

- It also occur in Membranous & Cartilage bone (Combination)

Ans-12)

Male

Skull :- Skull is the bony structure that protects the Cranial Cavity, & helps in form formation of structure of head.

- Skull is Composed of 22 bones
- Skull is used for identification like Age, Sex Race etc

- Male Skull is generally robust, large & more prominent then female skull

- Detail Discription of Male skull :-

1] General Appearance:

Male skull is larger, heavier & More - massive in nature

2) Architecture :-

Architecture of Male skull is Rugged or rough

3) Cranial Capacity :-

Cranial capacity of Male skull is 1400 - 1550 ML.

4] Digastric Groove :-

Digastric Groove of Male skull is deep whereas Female skull have shallow.

5] Parietal & Frontal Eminences :- Parietal & frontal Eminences of Male skull is less prominent, this is only characteristics which is less prominent in Male skull.

6] Mastoid Process :-

Large

7] Occipital Protoberance :-

Occipital Protoberance [At back] of male skull is More prominent.

8] Intra-nasal Angulation :-

It is angulated in male whereas

straight in female

9] Condylar facet :-
Condylar facet is large

10] Palatine :-
Palatine of Male skull is U-shaped

11] Zygomatic arch :-
Zygomatic arch [cheek bone] is more prominent in male skull

12] forehead :-
forehead of male skull is sloping whereas female has Vertical

13] Orbit :-
Orbital of male skull is rounded margin & Square

14] Foramen Magnum :-
It is long in nature

15] Nasal Aparture :-
Wide

16] chin :-
U-shaped

17] Lips :- large

Ans-
13.2 Somatometry :-

Somatometry is defined as measurement to various parts of the body.
It is used for the identification of human body.
Done by instruments- Anthropometer rod, sliding

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- forensic Significance :-

• Human Identification :-

Somatometry is used for the identification of human by measuring him/her prominent & whereas

• Sex :-

Male & female have different measurement like chest is broad then females, female pelvic wider than Male.

• Age :-

In determining age, we can use the of Somatometry by measuring height, weight

* Measuring process :- (for measurement we use landmarks :)

1] Head :-

Head length :- [g - op]

We measure from glabella [Medial point of Saggittal] to opisthocranion [back of skull]

- Head Circumference :- [g - op - g]

We measure from glabella - to opisthocranion then again glabella for finding overall measurement of head.

- Head Breadth :- [eu - eu]

We measure from Eurion [the lateral side of forhead] to Eurion [the other side]

Left to Right or vice - versa

2] Face :-

face measure is Generally divided into 2 type

- Morphological :- [n - gn]

We measure from nasion [Mid-point b/w eyes] (top of nose) to Gnathion [at chin-lower point]

- Physiological :- [t- gn]
  We measure from tryion [the uppermost point of head] to Gnathion.

3] ~~[illegible]~~ - Nose :-

Nose height :- [sn-pn]
We measure from sub-nasale [base of nose] to pro-nasale [tip of nose]

- Nose length :- [n- pn]
  nasion to pro-nasale

- Nose breadth :- [al to al]
  Measure from alare [the corner of nose] to alare

4] Height :-

Standing Height :-
Measure from Vertex (uppermost point on head) to floor which individual is in arm-hanging position with eye-ear

plane.

- Sitting height :- from Vertex to floor but with straight back with Eye - Ear plane.

Ans- 14.1 Parle Portrait / Bertilion System & Modern bio-metric Methods :- (4.5)

- Introduction :-

Bertilion System is broad catogorization of Somatoscopy & Somatometry method in which the individual [police] focuses on the characteristics of the person whereas Bio-metric Method is automatic method in which, the machine looks for person in its feeded record.

- ~~Comparison Description~~ Comparison :-

Bertilion system is classified into two types - Part A & Part B

Part A ↓ Descriptive Roll

Part B ↓ facial Recognition.

- Descriptive Role :- Bo Height, Build, Age, Sex, Skin complexion
- facial Recognition :-
  Lips, Head, forehead, Mouth, hair, Eye, Ear, Eyebrows, Nose, cheek, face etc.

- Bio-metric Methods of consist of finger print records of individuals, which matches the individuals with already feeded or old records to identify the person.

* Strengths & limitation Of Bertilion System & ~~Limitation~~ Biometric Method :-

- ~~Bio-method~~ Bio-metric Method give high accuracy then Bertilion system be cause fingerprint are unique in every individual whereas characteristics in broad Category are ~~not~~ difficult ~~unique~~ to recognize by person.
- Bio-metric Method is Easy & less time consuming whereas Bertilion System is time-Consuming because we have to identify Each characteristics.
- Bertilion System records are more safe whereas bio-metric method records get Easily manipulated

- Bertilion System is helpful when we recognize from distance we do not have susp criminal victim's fingerprint.
- Bio-metric method is Easily accessible then bertilion system.

Ans-

15.2 Facial Reconstruction :-

Facial Reconstruction is used to make the face for identification.
In this we use so many techniques.

* Various facial Reconstruction techniques :-

1] Man facial Repairing / Remodelling.
2] facial Superimposition Technique

* facial Superimposition techniques :-

1] 2- D- Photographic technique :- [Static]

In this technique we use a photo a person (suspect) on to skull for

identifying is this skull belong to this person.

2] Videographic technique :- [Motion]

In this technique we place

2D 3D.

- 2-D :-

We place the camera on both image & skull with same angle, distance, maximization for identification By this technique we can find different angles for identification.

- 3-D :-

We generally use this technique for finding the whole face from ear to ear by camera placement

- This technique is also done by software with blending, transparency, brightness and applying many more features.

* Facial Remodeling :-

We use Morphoscopy :- By visual appearance

Morphometric - measurement by tissue depth [placing tissue at different landmarks with appropriate thickness]

- By combining this two, we can repair face accarately.

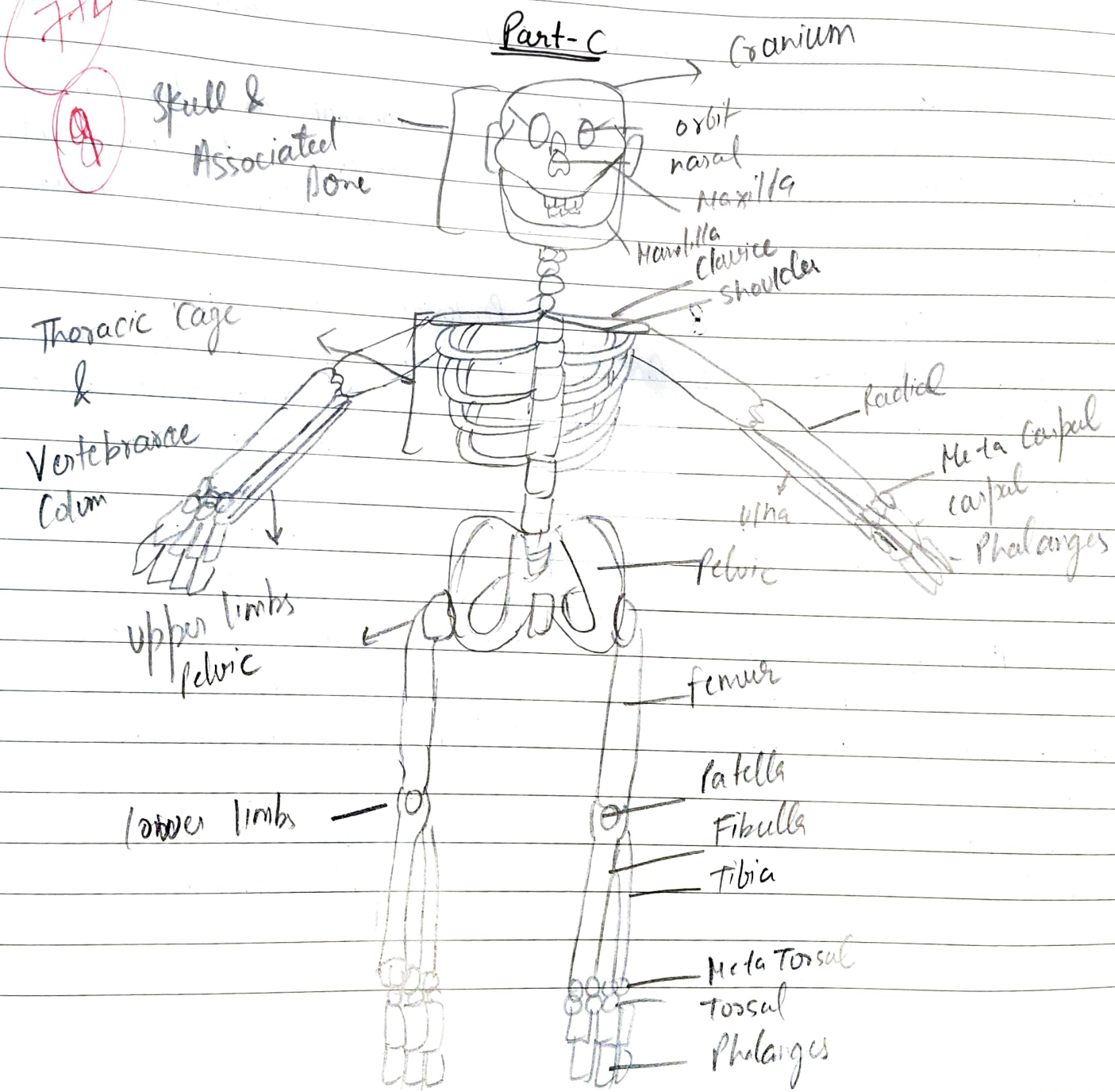
* Difference b/w manual & digital method :-

- Manual method is done by yourself with hands which is less accurate & needs Experts.
- Digital method is done by different software by using different tools. this method is more accurate.
- Digital method (softwares) is more Expensive than Manual method.
- Manual method can be done by clay & tissue or photographs.

Ans
-16.2

7+2

Part-C



Ans- 16:2

Skeleton is divided into 2 parts on the basis of region

- Axial skeleton
  Central structure of body

1] Skull & Associated Bones [28]
- Skull [22] → Cranium [8], face [14]
- Associated Bone [6]

+ Cranium [8]
- Parietal Bone [2] → Sides of head (Lateral)
- Occipital Bone [1] - At back of skull
- Temporal Bone [2] → Near Ear
- frontal Bone [1] - forehead
- sphenoid Bone [1] → At base neck [butter-fly shaped]
- Ethmoid Bone [1] → In ~~the~~ Eyes

+ face (14)
- Lacrimal [2] → ~~In~~ Near Eyes
- Vomer [1]
- Infra-nasal [2] → Nose
- Nasal [2] → Nose bone

Appendicular skeleton
Girdles & forelimbs which is responsible for movement

1] Pectoral Girdle [4]
- clavicle [2] → beauty bone
- Shoulder Giselle [2] shoulder bone

2] upper Limbs [60]
Humerus [2] - upper hand bone
Radius [2] - Thumb side bone
Ulna [2] - Pinky finger side bone
MetaCarpals (16) - Wrist bone
Carpals [10] - Palm bone
Phalanges [28] - finger bone

3] Pelvic bone [2]
Ilium, Ischium

Zygomatic bone [2] cheek bone
Palatine [1] → upper side of Mouth
Maxilla
Orbit [2] → upper part of Mouth
Mandible [1] → Lower jaw
Associated Bone [6]

† Ear
La Malleous - 2
stapes - 2
Lacus - 2

2] Thoracic Cage [25]
- Sacro sternum - 1 - chest bone
- Ribs [24 (1-12 pair)
- True Ribs - Direct Attached to sternum (1-7)

(7-11) false Ribs - Indirect Attach to sternum
(11-12) floating Ribs - Not Attach to sternum

3] Vertebrae Column [26]
Sacrum [1] Thoracic Vertebrae [12]
Coccyx [1]
Lomer [5]
Cervical [7]

4] Lower Limbs [60]

femur [2] - longest bone
Patella [2] - knee cap
Tibia [2] - lateral side
febulla [2] - Right side
Meta Torsal [14] - Ankle bone
Meto Torsals [10] - foot
Phalanges [28] - finger

Ans-

17.1 Skull :- Skull is the bony structure that protects the Cranial capacity of head & helps in forming the structure of head.

- Skull is composed of 22 bones.
- Skull is most reliable method for determining Age, Sex, Race for identification.

* Comparison chart :-

General characteristics features	Male skull	female skull
General Appearance	Large, Heavier, More Massive	Small, Lighter, Less Massive
Architecture	Rugged	Smooth
Wall & Surface	Large & Rough	Small & Smooth

	Male	female
Cranial Capicity	1400 - 1550 ml	1300 - 1450 ml
Occipital Occibrotuberance	More prominent	Less prominent
frontal & Parietal Eminence	less prominent	More prominent
Digastric Groove	Deep	Shallow
Palatine	U- shaped	V- shaped
Condylar fdce	Long	small
foramen Magnum	Long	short
Teeth	Large	Small
Chin	U- shaped	V- shaped

Forehead	Slight Sloping	Vertical
Nasal Aparture	Wide	Narrow
Intra- Nasal Angulation	Angulated	Straight
Mastoid Broces	Long	short
Orbit	Square, Round margin	Rough, sharp- margins
Zygomatic Arch	More Prominent	less prominent
Mouth	Large	Small

- for Determination of Sex of individual by skull we observe their morphological characteristics [by visual / Manually or digitally] like skull size, chin, zygomatic Arch etc & then note Each of the characterstic & Compare it with standard comparison chart [Difference chart].
- If The characteristics are more matches with male then it is of male skull & if it matches with female skull then id of female skull
- By Determining Sex, we can identify :-
  - Identify the Individual
  - Shorted out the suspects
  - Helps in forensic archaeology
  - Helps in forensic reconstruction
  - Helps in mass-disaster Cases.

Ans-

20.1 forensic Odontdogy :-

forensic Odontology is referred as Evaluation, Examination & identification of dental teeth for identification.

- Role of forensic Odontdogy in mass disaster cases :-

- Firstly Investigator divided into :-
  Home - Identification :- Collect anti-mortem records
  Site - Identification :- Examine dead- bodies.

- In Mass disaster cases generally the bodies are decomposed, burned etc
  So we try to find Andentity by :-

1] Age :-
  By tooth wear (pattern), Eruption sequence, accusal attribution [due to grinding] we can Estimate the age which is use for identification for person.

2] Sex :-
  Normally Male teeths are largest than

female, which indicates the sex of the individual.

3] Race:-

Due to Genetic difference, Environmental factor, teeth differentiate by Race which is used for narrow down the identification.

4] In Individual:-

~~Missing~~ Dental Anomalies:- Missing teeth / Extra teeth helps us to differentiate uniqueness in individual.

- Medical Record :-

If the individual have tooth cap then there is Serial no, or the doctor can recognize his work without an identity.

- DNA :-

By the use of ~~d~~teeth we can find the DNA by using polymerase chain Reaction.

- CAPMI :-

computerized Post-mortem Identification.

It is the machine which we used for ~~identi~~ Comparing the anti-mortem & post-mortem record, high accuracy

- Dental records are used for accurate Victim Identification when other methods fails ~~by~~ because :-
- Tooth (Enamel) are the most hardest ~~fo~~ bone in the our body in the case of mass-disaster Earthquake, it survives & get found Exact.
- We can used teeth for identification by finding all these characteristics described above.
- In Mass-disaster cases all the tissues are destroyed, burned & tears only the teeth are left for identification.
- They are highly used for age determination by Eruption sequence
- By Dental Anomalies, we can determine the uniqueness of individual.
- Dental records gene us the DNA by polymeran chain reaction which is used for identification.
- They used for in historical sites also for identification.

Ans- 18.1

8

Somatometry :-

Somatometry is defined as measurement of various parts of body.
It is generally used for identification of body specially personal identification.

Done by instruments :- Anthropometer rod, sliding calipers etc

- Different somatometric measurement :-

i] Head :-

Head length [g - op]

We measure the head length from glabella [Medial point of sagittal plane] to opisthocranion [at the back of skull]

- Head circumference [g - op - g]

We measure head circumference from glabella to opisthocranion and then again till glabella in single plane for finding overall measurement of head.

- Head Breadth :- [eu — eu]

We measure the head breadth from Eurycon [the lateral side of forehead] to Eurycon [the other lateral side of forehead].

2] Face :-

face measurement is divided into two types

- Morphological [n- Gn] height
  We measure the face length from nasion [top of nose] to Gnathion [at lower point of chin]

- Physiological [t- Gn]
  We measure the face height from t to Gnathion

3] Nose :-

Nose Height [sn-pn]
We measure the nose height from sub- nasion [base of nose] to pro- nasion [tip of nose]

- Nose length [n- prn]
We measure the nose length from nasion to pro nasion [tip of nose]

- Nose breadth [al -al]
We measure the nose breadth from alare [the corner of nose] to nose [alare].

4] Height :-
Standing Height :-
Measure from vertex [the uppermost point of head] to floor when the individual is standing in arm- hanging position with Eye- Ear plane.

- Sitting height :-
from vertex to floor but with straight back & Eye - Ear plane.

3] body weight :-
Measure from weighing machine [standing on weighing machine] with minimus clothes & bare foot

* Importance of Somatometry in forensic Examinations across diverse populations :-

1] Somatometry is used for human identification :-

We can determine age, height, sex, weight by measurement of human body.

2] Archaeology :-

We can determine old bodies by measurements of body.

3] Mass- disaster Case :-

We can use it in forensic identification of mass- disaster cases when the body is burned, decomposed etc.

4] Uniqueness :-

Every individual have unique characteristics which can be used for identfications.

(8)

Ans-

19.] facial Superimposition technique :-

facial Superimposition technique is used for identify the person, when the person is missing, burned, decomposed.

- facial superimposition technique :-

- Manual Method :- [Photographic]

We superimpose the picture or video onto a skull for identification manually, it needs Experts and have less accuracy.

- ~~2-D~~ Video-graphic Method :-

We ~~super~~ set up the camera on both skull & video then superimpose them for finding similarities it can be in 2-dimensional & 3-dimention.

- Digital Method :-

Both the photographic & Video-graphic method

Can be done by any software or digital machine by blending, sweeping, adjusting transparency & opaqueness, brightness.

- This method is more accurate but it is expensive.

- Strength & Limitation of facial superimposition techniques

- If we can find the individual by this technique.

- But manually there is less chance for accuracy.

- In facial superimposition techniques there is very difficult for accurate tissue depth on specific landmark which can ~~distorb~~ destroy the whole face

- facial superimposition technique software is expensive & need high-Expertise man in both manually & digitally.

- facial superimposition technique needs high skills & patience

- If the person is tissue is placed at correct place in half we can able to determine person.

- Confirmed & disputed Identity Cases :-

- When the picture or person is confirmed we can easily identify the person.
- But in disputed normally the tissue get disturbed & distributed many time which can lead so many wrong leads.
- In ~~confirmed~~ ~~confirmed~~ identity cases, the digital method find the person by their old record which is little easier than ~~to~~ ~~confirmed~~ disputed identity case.
- In disputed cases, we get very less idea of person by leftour pieces but in confirmed identity some time we get full idea or zero idea of person by face records.
- In confirmed case, & disputed case facial superimposition technique limitation & strength depends on the condition of person.

End of Writing